KUMAR RAJA PUVVALLA

Platform & Infrastructure Engineer

Location: Glasgow • **Contact:** +44 7918392150 • **Email:** kumararajapuvvalla@gmail.com • Linkedin

PROFILE SUMMARY

Platform Engineer with 3+ years of hands-on experience building and operating production grade cloud infrastructure across AWS, GCP, and Azure. Specialises in Kubernetes cluster management, Terraform IaC, CI/CD automation, and backend service development that abstracts infrastructure complexity for engineering teams. Proven track record designing self service platforms that improve developer experience, reduce deployment failures, and drive operational excellence at scale. Comfortable working across both platform engineering and software development from Kubernetes controllers and backend APIs to multi-AZ cluster architecture and service mesh configuration.

CORE COMPETENCY

Cloud Infrastructure • Kubernetes Platform Engineering • CI/CD Pipeline Design & Automation • Infrastructure as Code • Service Mesh & Networking • Backend Service Development • Platform Observability • Self Service Platforms • Cloud Security & DevSecOps

IT SKILLS

AWS (EKS, EC2, VPC, IAM, ALB, RDS, S3, Lambda, CloudTrail, AWS Config) • GCP (Cloud Run, Cloud SQL, Cloud Functions, BigQuery) • Azure (AKS, VNet, Azure AD, Key Vault, Monitor, Log Analytics, Databricks) • GitHub Actions • Azure DevOps • Jenkins • Terraform • ARM • CloudFormation • Ansible • Kubernetes (EKS, AKS) • Helm • Argo CD • Docker • Envoy Proxy • HPA • VMSS Node Pools • Linux • Secrets Manager •Key Vault • SonarQube • Semgrep • CloudWatch • Prometheus • Grafana • ELK • Bash • Python • YAML• Go

WORK EXPERIENCE

Tech Nxt Gen, UK, as Platform Engineer (Hybrid working) **Dec 2024 – Present**

Job Profile: *Platform engineering lead on a Cloud Native AI SaaS product responsible for the full AWS/EKS production platform, CI/CD pipelines, and operational reliability of containerised workloads.*

- Architected and operate a production AWS EKS platform across multi-AZ clusters with ALB ingress, HPA auto scaling, and Terraform managed IRSA roles, sustaining 99.9% uptime.
- Designed VPC networking with layered public/private subnets, NAT gateways, and security group isolation, providing controlled workload separation across environments.
- Built end to end CI/CD pipelines in GitHub Actions with integrated SonarQube and Semgrep security scanning and automated Helm chart validation, reducing release lead time by 45%.
- Strengthened platform security posture using IAM least-privilege policies, CloudTrail audit logging, and AWS Config compliance rules — resolved a critical audit finding within 9 days.
- Reduced AWS cloud spend by 10% through instance rightsizing, S3 lifecycle policies, and automated zombie resource cleanup via tagging enforcement.
- Rationalised observability from 85 noisy alerts per week to 11 actionable signals with structured CloudWatch severity tiers, cutting mean time to acknowledge from 22 to 4 minutes.
- Developed Go based internal tooling to automate cluster lifecycle tasks including node drain sequencing and stale resource cleanup across EKS environments.
- Resolved complex Kubernetes production incidents including GPU pod OOM, cluster API latency from accumulated batch job history, and OIDC authentication drift across all delivery pipelines.

PWC Information Technology, UK, as Cloud Platform Engineer (Remote) **May 2023 – Nov 2024**

Job Profile: *Delivered cloud platform infrastructure and CI/CD automation for a major UK bank migrating from on-premise legacy systems to Azure at scale.*

- Led Terraform infrastructure provisioning across multiple Azure subscriptions introduced reusable modules and provider version pinning after a P0 incident and governance standard adopted across all delivery teams.
- Designed and standardized a CI/CD pipeline framework across 6 delivery teams using Azure DevOps and GitHub Actions with reusable YAML templates and environment promotion gates, reducing deployment failures by 30%.
- Replaced a static 12 agent pool with a Terraform managed Azure VMSS auto scaling fleet (4 idle, 40 at peak) and implemented shared Docker layer caching, cutting pipeline queue time from 140 to under 12 minutes on release days.

- Migrated all Azure DevOps Service Connections to Workload Identity Federation (OIDC), eliminating credential rotation failures and achieving zero authentication pipeline incidents over 9 months.
- Rebuilt monitoring and observability strategy in Azure Monitor and Log Analytics consolidated 340 alerts into 142 actionable signals and improved incident response time by 40%.
- Provisioned and automated Azure Databricks, AKS, Azure SQL, Storage Accounts, and Data Factory, improving developer self service onboarding speed by 26% through standardized environment automation.
- Developed Ansible playbooks for automated server provisioning and maintained a Jenkins CI ecosystem with Nexus repository management supporting microservices on Azure.

Elekta, UK, As a DevOps Engineer (Hybrid) **Aug 2022 – Apr 2023**

Job Profile: *Supported cloud infrastructure and CI/CD automation for Elekta's medical technology platforms, working across AWS and Azure in a regulated healthcare environment.*

- Built and maintained CI/CD pipelines using Jenkins and Azure DevOps to automate build, test, and deployment workflows, reducing manual steps and improving release reliability for oncology platform components.
- Provisioned and managed cloud infrastructure on AWS and Azure using Terraform and CloudFormation, supporting secure and repeatable environment deployments for medical device software.
- Containerised application components using Docker and deployed them to Kubernetes clusters using Helm, improving deployment consistency across development and staging environments.
- Developed Ansible playbooks to automate server configuration and backend environment setup, reducing environment provisioning time and minimising configuration drift.
- Administered Linux (Ubuntu) systems including patching coordination, performance monitoring, and deployment support across development and test environments.
- Implemented CloudWatch monitoring and alerting to improve system visibility and support proactive incident detection across AWS hosted platform services.

PROJECTS

Infrastructure as Code Adoption | Terraform · CloudFormation · AWS · Azure · Kubernetes · Helm · Git | Sep 2022 – Apr 2023

- Migrated manual environment setup to Terraform managed AWS and Azure infrastructure built reusable modules for VPC, Kubernetes clusters, and database tiers enabling repeatable deployments across dev, staging, and production environments.
- Reduced provisioning time and eliminated configuration drift by enforcing peer reviewed, version controlled infrastructure changes.

Pipeline Standardisation & Agent Autoscaling | Azure DevOps · Terraform · VMSS · Docker · AKS · GitHub Actions | May 2023 – Nov 2024

- Designed and built a standardised CI/CD pipeline framework across 6 delivery teams reusable YAML templates, environment promotion gates, and Workload Identity Federation replacing all client secret authentication; reduced deployment failures and eliminated credential rotation outages entirely.
- Replaced a static 12 agent pool with a Terraform managed Azure VMSS auto scaling agent fleet (4 idle, scales to 40 at peak) and implemented shared Docker layer caching via Azure Blob Storage, cutting pipeline queue time from 140 minutes to under 20 minutes on release days.

Production EKS Platform - AI SaaS| AWS · Terraform · GitHub Actions · Helm · ArgoCD · CloudWatch · IAM · SonarQube | Jan 2025 - Present

- Designed and provisioned the full AWS production platform from scratch using Terraform. Multi AZ VPC with public/private subnets, EKS cluster with ALB ingress, HPA auto scaling, and IAM least privilege IRSA roles per workload established GitOps delivery via ArgoCD with Helm chart versioning and canary rollout strategy.
- Built the end to end CI/CD pipeline in GitHub Actions with integrated security scanning (SonarQube, Semgrep), automated Helm chart validation, and environment promotion gates reducing release lead time and enabling zero downtime deployments to production.

Insurance Cloud Platform - GCP/Azure Hybrid (Personal Project) In Progress | Jan 2026 – Present

Stack: GCP · Azure DevOps · Terraform · Cloud Run · Cloud SQL · Dialogflow CX · BigQuery · Cloud Functions · Python · FastAPI · Docker

- Designing and building a full insurance company backend on GCP from scratch. VPC with layered public/private/SQL subnets, default deny firewall rules, Cloud NAT, a Cloud Run REST API connecting to a private Cloud SQL database via VPC Access Connector, all credentials injected at runtime from Secret Manager with no secrets in code.
- Backend REST API built in Python/FastAPI with Go based Cloud Functions for event driven ETL processing.
- Deploying the complete stack via an Azure DevOps pipeline with canary deployment integrated a Dialogflow CX chatbot for customer policy and claims queries, a serverless ETL pipeline via Cloud Functions and Cloud Workflows, and BigQuery for daily reporting.

EDUCATION

- Bachelor's in computer science, JNTU Hyderabad - INDIA
- Master's in computer science, University of Hertfordshire - UK